

AI Trust & Operations — Master Document

Overview

This document consolidates **all phases (1 → 6)** of the AI Trust & Operations system into a single, enterprise-grade reference. It is designed to be: - Downloadable - Auditable - Executable as a real system blueprint

1. Business Value & Why This System Exists

Goal: Create a regulated, explainable, human-governed AI Trust Operating System.

Problems Solved

- Silent AI drift
- Unexplainable decisions
- Over-automation risk
- Regulatory exposure
- Loss of human trust

Value

- Faster decisions with human safety
 - Audit-ready AI
 - Scalable trust governance
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2. Architecture Overview (Component + Data Flow)

Core Layers

1. Trust Data Layer (TRS + TRUST-OPS-010)
2. AI Intelligence Layer (AI-TRUST-001/002/003)
3. Governance & Oversight Layer (AI-OPS-004/005/006)

High-Level Data Flow

Seller Event → Trust Ops → AI Risk → Behavior Analysis → Decision Engine → Human Review → Feedback → Governance

3. Technical Deep Dive (Module by Module)

AI-TRUST-001 — Risk Scoring

- Deterministic weighted model
- Inputs: Trust score, escrow, velocity
- Output: riskScore (0-1), explainability

AI-TRUST-002 — Behavior & Fraud Patterns

- Temporal analysis (24h → 90d)
- Pattern detection (refunds, spikes, volatility)
- Output: behaviorRiskScore + patterns

AI-TRUST-003 — Decision Engine

- Synthesizes AI + trust + context
- Outputs recommendations only
- Escalation levels (NONE → ESCALATE)

AI-OPS-004 — Monitoring & Drift

- Drift detection
- Accuracy tracking
- Health scoring

AI-OPS-005 — Human Feedback Loop

- Human override capture
- Alignment metrics
- Improvement signals (read-only)

AI-OPS-006 — Change & Rollout Control

- Shadow evaluation
- Human approval
- Rollback & versioning

4. Governance, Compliance & Explainability

Principles

- Humans always decide
- AI never auto-enforces
- Everything explainable

Audit Coverage

- Every decision
 - Every override
 - Every model change
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5. Operating Model (Who Does What)

Role	Responsibility
Analyst	Review & feedback
Supervisor	Approve changes
Manager	Governance oversight
Admin	Emergency rollback

6. Production Readiness & Rollout

Deployment Strategy

- Shadow first
- Human approval
- Controlled rollout

KPIs

- AI Health Score
 - Human trust score
 - Override rate
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Final Statement

This system is not just AI. It is a **regulated AI Trust Operating System** designed for real companies, real money, and real risk.